MIL-C-46389A(MU)
14 February 1964
SUPERSEDING
MIL-C-46389(MU)
1 August 1963

MILITARY SPECIFICATION

CORRECTOR, CANT: XM4

1. SCOPE

1.1 This specification covers one type of instrument known as Cant Corrector: XM4, primarily intended for mounting on the 105MM Howitzer XM102.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45208	Inspection Requirements, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F8626159	Corrector, Cant: XM4
SQAP 8626161	Supplementary Quality Assurance Provisions for Level 8626161

FSC 1290

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PACKAGING DATA SHEET

MIL-P-14232/P8626159 Packaging of Corrector, Cant: XM4

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The cant corrector shall be manufactured in accordance with Drawing F8626159 "Corrector, Cant: XM4" and drawings pertaining thereto

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Operating temperatures.- The cant corrector shall meet the applicable requirements of 3.5 to 3.5.4 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.4.2 Vibration.- With the cant corrector locked against rotation, the cant corrector shall be vibrated at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes ± fifteen seconds. Subsequent to vibration the lock shall have remained engaged and there shall be no evidence of damage or physical failure. With the lock disengaged the cant corrector shall meet the requirements of 3.5 to 3.5.3 inclusive.

3.5 Orientation.- The cant corrector shall be mounted on a horizontal surface, the cross level vial centered and the plane of the gunner's quadrant pad horizontal within 0.10 mil.

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3.5.1 Quadrant pads.- The plane of the vertical shoulder adjacent to the gunner quadrant pads shall be parallel to the axis of rotation within 1.0 mil.

3.5.2 Rotation.- With the rotational lock disengaged and the cant corrector stabilized at an ambient temperature of 72°F, the adjusting screws of the brackets shall be adjusted so that there is free rotation (torque not to exceed 8 ounce inches) of the housing containing the level vial. The housing shall be operable (rotate) at extreme temperatures of 3.4.1.

3.5.3 Excursion.- The housing section containing the level vial shall move through an excursion of at least 230 mils in one direction and 1383 mils in the opposite direction from the position established in 3.5 when the rotation lock is disengaged.

3.5.4 Lock.- The rotational lock, when engaged, shall prevent rotation of the housing section.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.-

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Presentation of Lots" and "Drawing of Samples" specified in Standard MIL-STD-105.

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4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and Supplementary Quality Assurance Provisions, SQAP-8626161. Examinations and tests related to Section 3 therein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Tables I and II titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

CRITICAL: NONE

MAJOR: AQL 1.0% Defective		REQUIREMENTS	TEST PROCEDURE
101.	Physical failure - post vibration	3.4.2	4.5.3
102.	Quadrant pad housing misaligned	3.5.1	4.5.4
103.	Rotation	3.5.2	4.5.6
104.	Excursion	3.5.3	4.5.7
105.	Lock	3.5.4	4.5.5

MINOR: NONE

NOTE: Above tests to be conducted at standard ambient temperature (60° to 90°F)

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level S4 in Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

CRITICAL: NONE

MAJOR: AQL 1.5% Defective		REQUIREMENTS	TEST PROCEDURE
101.	Operability failure of the corrector at the extreme temperature	3.4.1	4.5.2

MINOR: NONE

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4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Disposition of Non-conforming Product" specified in Standard MIL-STD-105.

4.3 Control tests.- One corrector shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Materials	3.1	4.6.1
Fabrication	3.2	4.6.1
General Specifications	3.3	Visual-MIL-F-13926

4.3.1 Control test failure.- Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL-I-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment" and Government Furnished Material" requirements of Specification MIL-I-45208.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL-I-45208.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

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4.5 Test methods and procedures.

4.5.1 Fabrication and materials.- To determine compliance with 3.1 and 3.2, the corrector shall be visually inspected for proper materials and fabrication in accordance with the specified drawings, and drawings pertaining thereto. Dimensional checks shall be made by use of standard measuring equipment. This inspection shall be performed on the control sample (4.3).

4.5.2 Operating temperature.- The operating temperature test consists of exposing the corrector to the temperatures indicated in 3.4.1. The test equipment shall conform to the "Test Facilities" requirements of Specification MIL-F-13926, and the conditions of 3.4.1 shall apply.

4.5.3 Vibration.- Standard vibration equipment capable of providing the amplitude and frequency while vibrating in a vertical plane shall be utilized in conformance with the conditions specified in 3.4.2. Unless otherwise specified, the test equipment shall meet the "Test Facilities" requirements of Specification MIL-F-13926. The mounting surface of the corrector shall be in a horizontal plane within two (2) degrees when secured to the vibrating equipment and the locking mechanism of the corrector in the locked position.

4.5.3.1 Post vibration.- After removing the corrector from the vibration equipment, the position of the level vial bubble shall not exceed the requirements as specified in 3.5. The corrector shall be thoroughly examined for loose, damaged, or missing parts resulting from poor workmanship and materials, to determine compliance with 3.4.2 prior to being subjected to the remaining performance tests. A visual examination shall be made for missing or damaged parts such as the level vial assembly and the"push to lock" mechanism. A smell test shall be used to detect a leaking level vial resulting from a broken seal or glass. A tactile examination shall be made to determine excessive looseness or improper function of such moving parts as rotation of the member supporting the level vial when the assembly is unlocked, the level vial cover, and the push action of the knobs to lock and unlock the assembly, as well as screws, retainers, and other assembled parts which may have loosened as a result of vibration. Any one of the possible defects defined above shall be cause to consider a corrector as defective unless separate defects-per-hundred type sampling plans are established and approved by the Government as part of the suppliers inspection system. Correctors which have passed the above examinations shall be subjected to the tests of 4.5.4 to 4.5.7 inclusive to determine overall conformance with 3.4.2.

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4.5.4 Quadrant pads.- With the corrector positioned as specified in 3.5.1 the plane of the vertical shoulder shall be parallel to the axis of rotation within the requirement specified in 3.5.1 when checked with a precision 10-second level.

4.5.5 Lock.- This is a test to determine conformance with 3.5.4 for rotational lock engagement. When the lock is engaged, the housing section shall not rotate when finger pressure is applied.

4.5.6 Rotation.- This test shall consist of measuring the torque required to rotate the corrector housing through full travel at standard ambient temperature by means of a torque wrench with an adapter capable of accepting the housing to determine compliance with 3.5.2. Subsequent to the torque readings, the housing shall be rotated at the temperatures of 3.5.1 to insure operability in conformance with 3.5.2.

4.5.7 Excursion.- The housing section containing the level vial shall be rotated to the mil values and directions specified in 3.5.3 after disengaging the rotation lock. The corrector shall be positioned in accordance with 4.5.4 prior to performing this test.

4.5.8 Packaging, packing and marking.- Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8626159. The level of protection shall be specified in the procurement document.

6. NOTES

6.1 Intended use.- The Cant Corrector XM4 is primarily intended for mounting on the gun carriage of the 105MM Howitzer XM102 to provide a surface that is horizontal in the direction of the line of fire thus providing a mounting surface for a gunner's quadrant to check the angle of cant of the gun trunnion regardless of the elevation or depression of the gun tube without introducing level vial roll into the quadrant.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

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